Document containing data related to statistical analyses for the work: *Behavioral Disruption in Honey Bees (Apis mellifera) Exposed to Isolated and Combined Insecticides* : Ecotoxicology Journal

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## 1 Walking speed

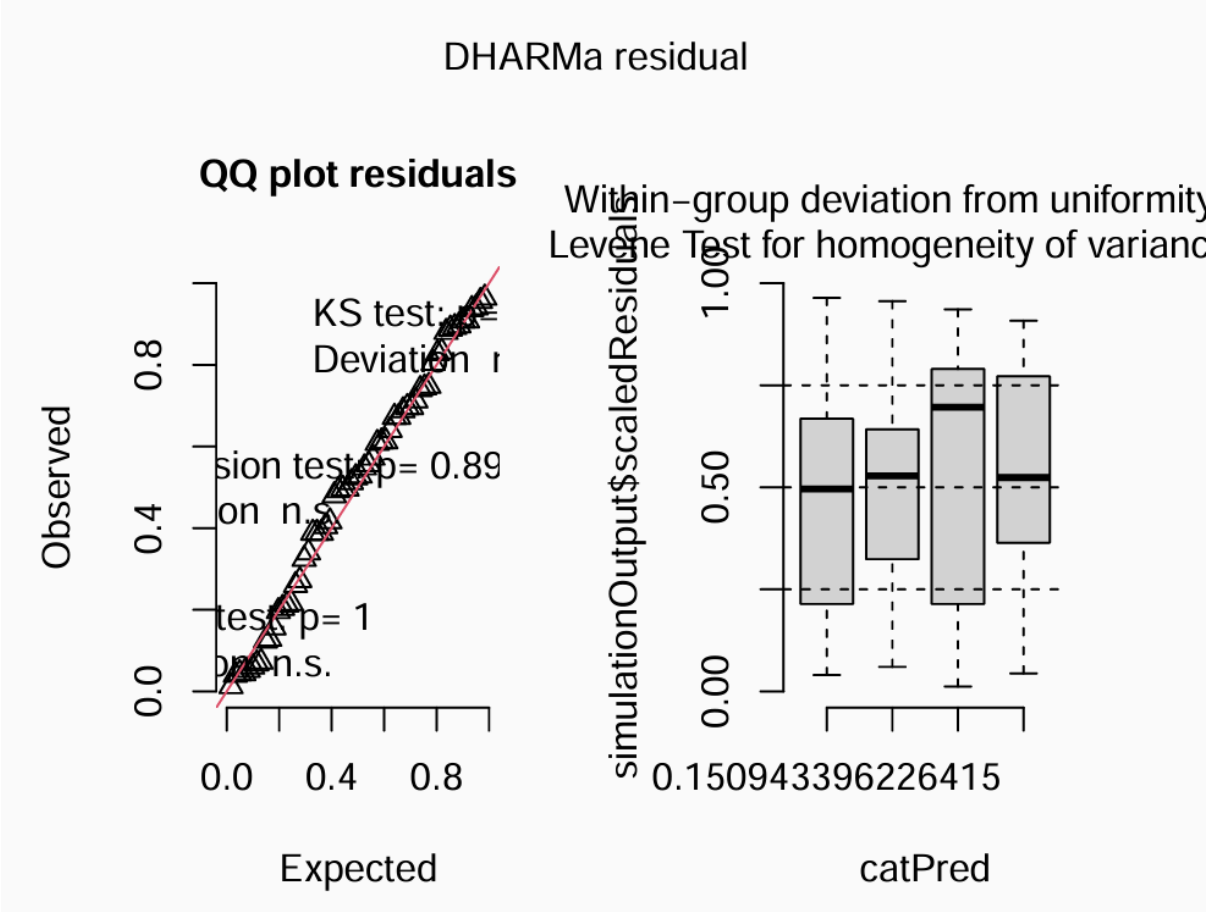
1.1 Descriptive table with means (cm/s), sample size (n), standard deviation (SD), and standard error (SE) for walking speed.

| Group | Mean | n | SD | SE |
|---|---|---|---|---|
| CTR | 3.57 | 15 | 1.27 | 0.329 |
| DEL | 3.01 | 15 | 1.39 | 0.360 |
| IMI | 2.54 | 15 | 1.15 | 0.298 |
| COM | 2.79 | 15 | 1.00 | 0.258 |

1.2 Model selection tables based on Akaike Information Criteria (AIC) for walking speed

| Mod_name | Family | Random | df | AICc | Delta |
|---|---|---|---|---|---|
| mod_velocidade.3 | gaussian(id) | | 5 | 200.5 | 0.00 |
| mod_velocidade.4 | gaussian(id) | (1 \| video)) | 6 | 203.0 | 2.42 |
| mod_velocidade.2 | negative binomial (lg) | | 5 | 215.3 | 14.77 |
| mod_velocidade.1 | negative binomial (lg) | | 5 | 215.3 | 14.77 |

1.3 Best model residuals: mod_velocidade.3




1.4 Best model for walking seed — mod_velocidade.3 (gaussian, identity) — summary

| | **Estimate** | **std.error** | **z value** | **p value** |
|---|---|---|---|---|
| Intercept(CTR) | 3.5718 | 0.3029 | 11.793 | $2.10^{-16}$ *** |
| DEL | -0.5666 | 0.4283 | -1.323 | 0.1859 |
| IMI | -1.0357 | 0.4283 | -2.418 | 0.0156 * |
| COM | -0.7835 | 0.4283 | -1.829 | 0.0674 . |

## 2 Traveled distance

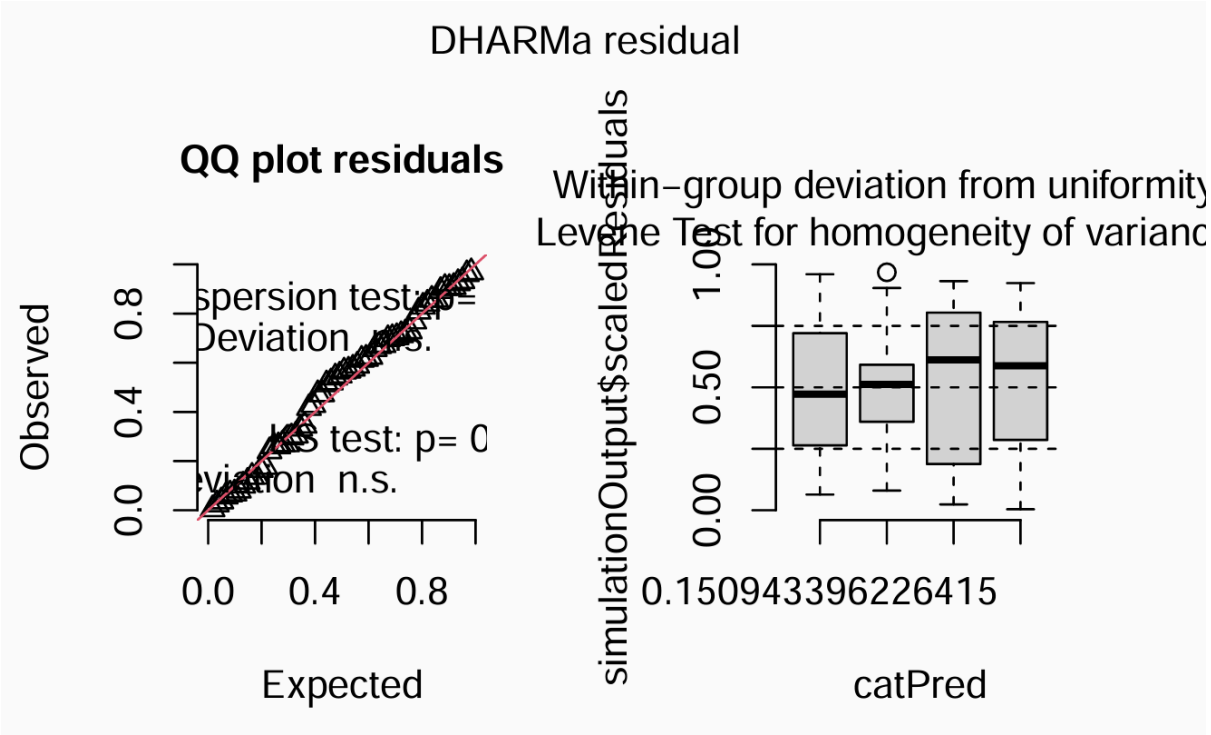
2.1 Descriptive table with means, sample size (n), standard deviation (SD), and standard error (SE) for traveled distance (cm).

| Group | Mean | n | SD | SE |
|---|---|---|---|---|
| CTR | 943.0859 | 15 | 459.5198 | 118.6475 |
| DEL | 844.3245 | 15 | 451.5633 | 116.5936 |
| IMI | 634.7754 | 15 | 371.0709 | 95.8101 |
| COM | 736.4233 | 15 | 332.3832 | 85.8210 |

2.2 Model selection tables based on Akaike Information Criteria (AIC) for traveled distance (cm)

| Mod_name | Family | Random | df | AICc | Delta |
|---|---|---|---|---|---|
| mod_distancia.3 | gaussian(id) | | 5 | 898.4 | 0.00 |
| mod_distancia.4 | gaussian(id) | (1\| Video) | 6 | 900.8 | 2.47 |
| mod_distancia.2 | negative binomial2(lg) | | 5 | 909.8 | 11.44 |
| mod_distancia.1 | negative binomial1(log) | | 5 | 910.7 | 12.29 |

2.3 Best model residuals: mod_distancia.3




1.4 Best model for walking seed — mod_distancia.3 (gaussian, identity) — summary

| | **Estimate** | **std.error** | **z value** | **p value** |
|---|---|---|---|---|
| Intercept(CTR) | 943.09 | 101.57 | 9.285 | $2.10^{-16}$ *** |
| DEL | -98.76 | 143.65 | -0.688 | 0.4917 |
| IMI | -308.31 | 143.65 | -2.146 | 0.0318 * |
| COM | -206.66 | 143.65 | -1.439 | 0.1502 |

**3 Moving time**

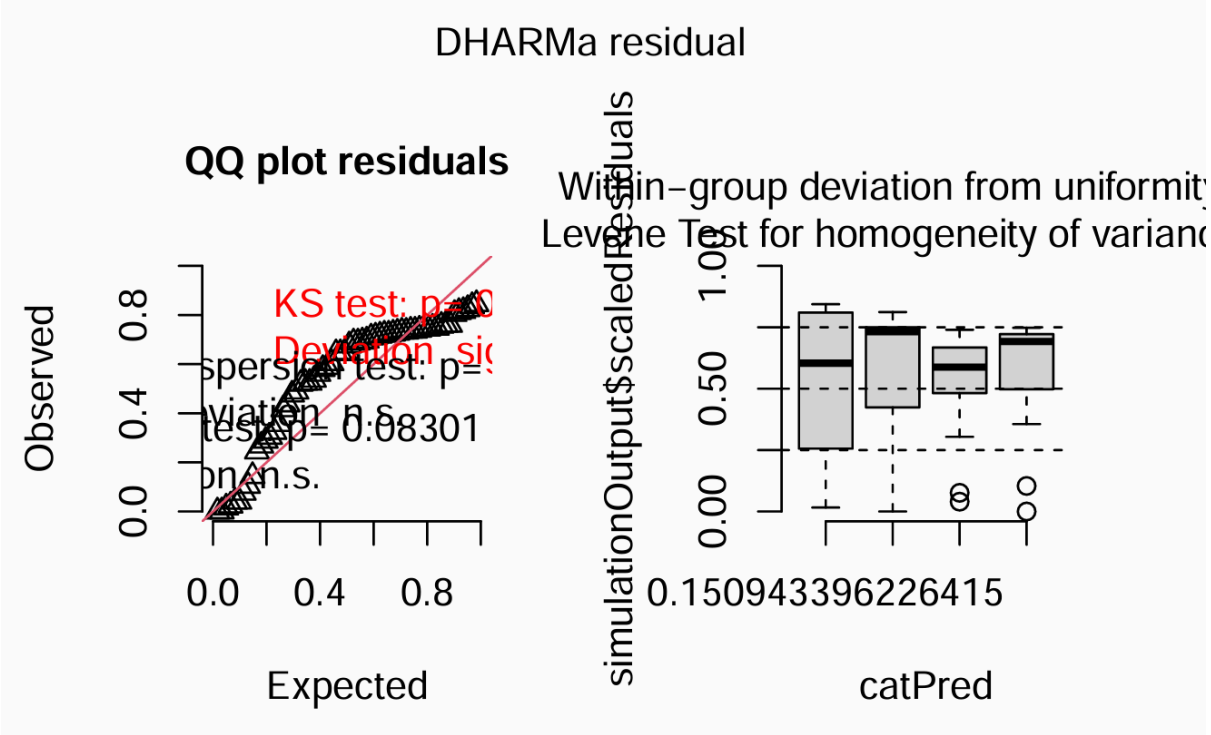
3.1 Descriptive table with means, sample size (n), standard deviation (SD), and standard error (SE) for moving time (seconds).

| **Group** | **Mean** | **n** | **SD** | **SE** |
|---|---|---|---|---|
| CTR | 249.9791 | 15 | 81.7790 | 21.1152 |
| DEL | 258.1860 | 15 | 79.7374 | 20.5881 |
| IMI | 231.8921 | 15 | 72.1941 | 18.6405 |
| COM | 256.8207 | 15 | 53.0675 | 13.7020 |

3.2 Model selection tables based on Akaike Information Criteria (AIC) for time moving (seconds)

| **Mod_name** | **Family** | **Random** | **df** | **AICc** | **Delta** |
|---|---|---|---|---|---|
| glm_moving.2 | gaussian(id) | | 5 | 691.4 | 0.00 |
| glm_moving.3 | gaussian(id) | (1\| Video) | 6 | 693.9 | 2.47 |
| glm_moving.1 | negative binomial 1(lg) | | 5 | 732.2 | 40.75 |

3.3 Best model residuals: glm_moving.2




3.4 Best model for walking seed — mod_distancia.3 (gaussian, identity) — summary

| | **Estimate** | **std.error** | **z value** | **p value** |
|---|---|---|---|---|
| Intercept(CTR) | 943.09 | 101.57 | 9.285 | $2.10^{-16}$ *** |
| DEL | -98.76 | 143.65 | -0.688 | 0.4917 |
| IMI | -308.31 | 143.65 | -2.146 | 0.0318 * |
| COM | -206.66 | 143.65 | -1.439 | 0.1502 |

## 4 Grooming

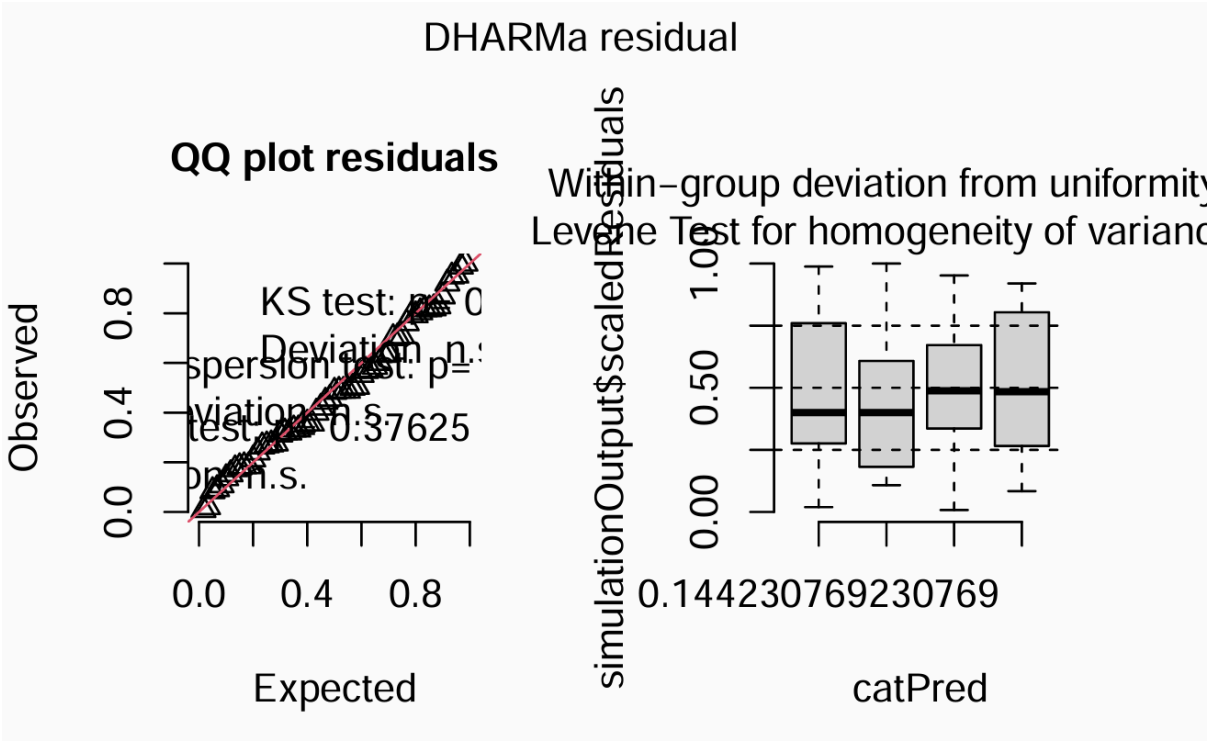
4.1 Descriptive table with means, sample size (n), standard deviation (SD), and standard error (SE) for grooming (seconds).

| Group | Mean | n | SD | SE |
|---|---|---|---|---|
| CTR | 8.9286 | 15 | 11.3033 | 2.9185 |
| DEL | 30.2000 | 15 | 58.2990 | 15.0526 |
| IMI | 73.2334 | 15 | 56.3139 | 14.5407 |
| COM | 70.8667 | 15 | 60.6902 | 15.6702 |

4.2 Model selection tables based on Akaike Information Criteria (AIC) for grooming (seconds)

| Mod_name | Family | Random | df | AICc | Delta |
|---|---|---|---|---|---|
| mod_limpeza.2 | tweedie (lg) | (1 \| Video) | 7 | 559.0 | 0.00 |
| mod_limpeza.1 | tweedie (lg) | | 6 | 559.8 | 0.78 |

4.3 Best model residuals: mod_limpeza.2




3.4 Best model for grooming — mod_limpeza.2 (tweedie, log) — summary

| | **Estimate (log)** | **std.error** | **z value** | **p value** |
|---|---|---|---|---|
| Intercept(CTR) | 2.1195 | 0.3060 | 6.927 | $4.30.10^{-12}$ *** |
| DEL | 1.0841 | 0.3802 | 2.851 | 0.00435 ** |
| IMI | 2.1566 | 0.3523 | 6.122 | $9.25.10^{-10}$ *** |
| COM | 2.0532 | 0.3501 | 5.864 | $4.51.10^{-09}$ *** |

**5 wing fanning**

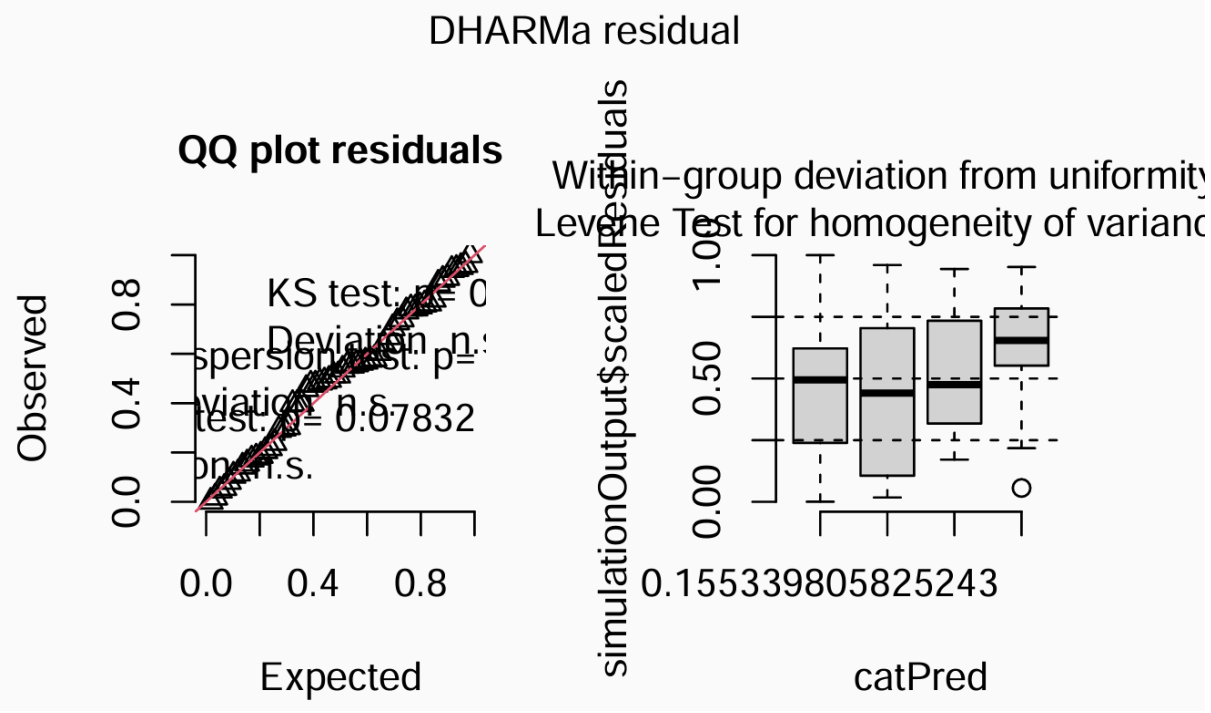
5.1 Descriptive table with means, sample size (n), standard deviation (SD), and standard error (SE) for wing fanning (seconds).

| Group | Mean | n | SD | SE |
|---|---|---|---|---|
| CTR | 32.7667 | 15 | 59.7575 | 15.4293 |
| DEL | 23.7667 | 15 | 76.4415 | 19.7371 |
| IMI | 24.0000 | 15 | 57.5837 | 14.8681 |
| COM | 37.9286 | 15 | 51.8021 | 13.3752 |

5.2 Model selection tables based on Akaike Information Criteria (AIC) for wing fanning (seconds)

| Mod_name | Family | Random | df | AICc | Delta |
|---|---|---|---|---|---|
| mod_wing.1 | tweedie (lg) | | 6 | 357.5 | 0.00 |
| mod_wing.2 | tweedia (lg) | (1 \| Video) | 7 | 360.0 | 2.51 |

5.3 Best model residuals: mod_wing.1




5.4 Best model for wing fanning — mod_wing.1 (tweedie, log) — summary

| | **Estimate (log)** | **std.error** | **z value** | **p value** |
|---|---|---|---|---|
| Intercept(CTR) | 3.4894 | 0.5438 | 6.416 | $1.4.10^{-10}$ *** |
| DEL | -0.3211 | 0.7926 | -0.405 | 0.685 |
| IMI | -0.3114 | 0.8066 | -0.386 | 0.699 |
| COM | 0.1463 | 0.7722 | 0.189 | 0.850 |

## 6 Syrup consumption

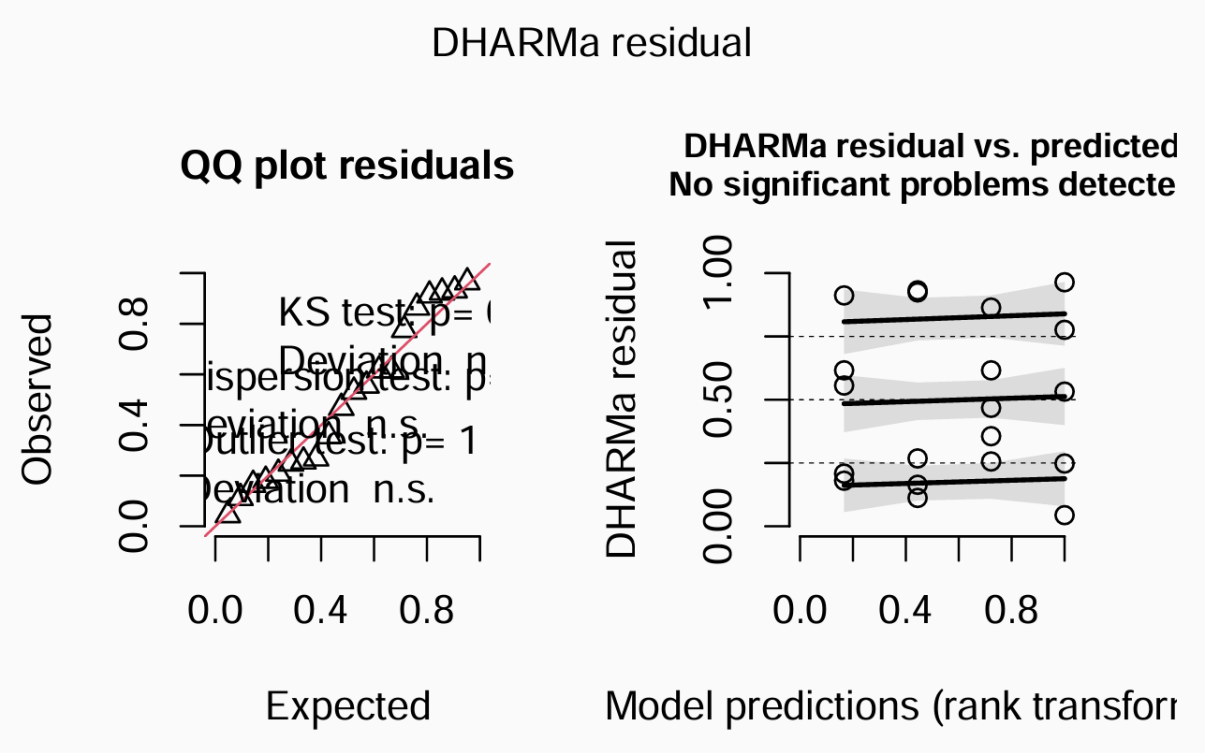
6.1 Descriptive table with means, sample size (n), standard deviation (SD), and standard error (SE) for syrup consumption (mg.24h$^{-1}$).

| Group | Mean | n | SD | SE |
|---|---|---|---|---|
| CTR | 32.5220 | 5 | 13.6586 | 6.1083 |
| DEL | 42.3057 | 5 | 28.6427 | 12.8094 |
| IMI | 19.3675 | 5 | 17.9090 | 8.0092 |
| COM | 24.8963 | 5 | 25.6773 | 11.4833 |

6.2 Model selection tables based on Akaike Information Criteria (AIC) for syrup consumption (mg.24h$^{-1}$)

| Mod_name | Family | Random | df | AICc | Delta |
|---|---|---|---|---|---|
| mod_consumo.g | gaussian | | 5 | 190.7 | 0.00 |
| mod_consumo.g.1 | gaussian | (1 \|rep) | 6 | 194.9 | 4.12 |

6.3 Best model residuals: mod_consumo.g




6.4 Best model for syrup consumption — mod_consumo.g (gaussian, identity) — summary

| | **Estimate (log)** | **std.error** | **z value** | **p value** |
|---|---|---|---|---|
| Intercept(CTR) | 32.522 | 8.915 | 3.648 | 0.000264 *** |
| DEL | 9.784 | 12.608 | 0.776 | 0.437752 |
| IMI | -13.155 | 12.608 | -1.043 | 0.2968 |
| COM | -7.626 | 12.608 | -0.605 | 0.5453 |

## 7 Mortality

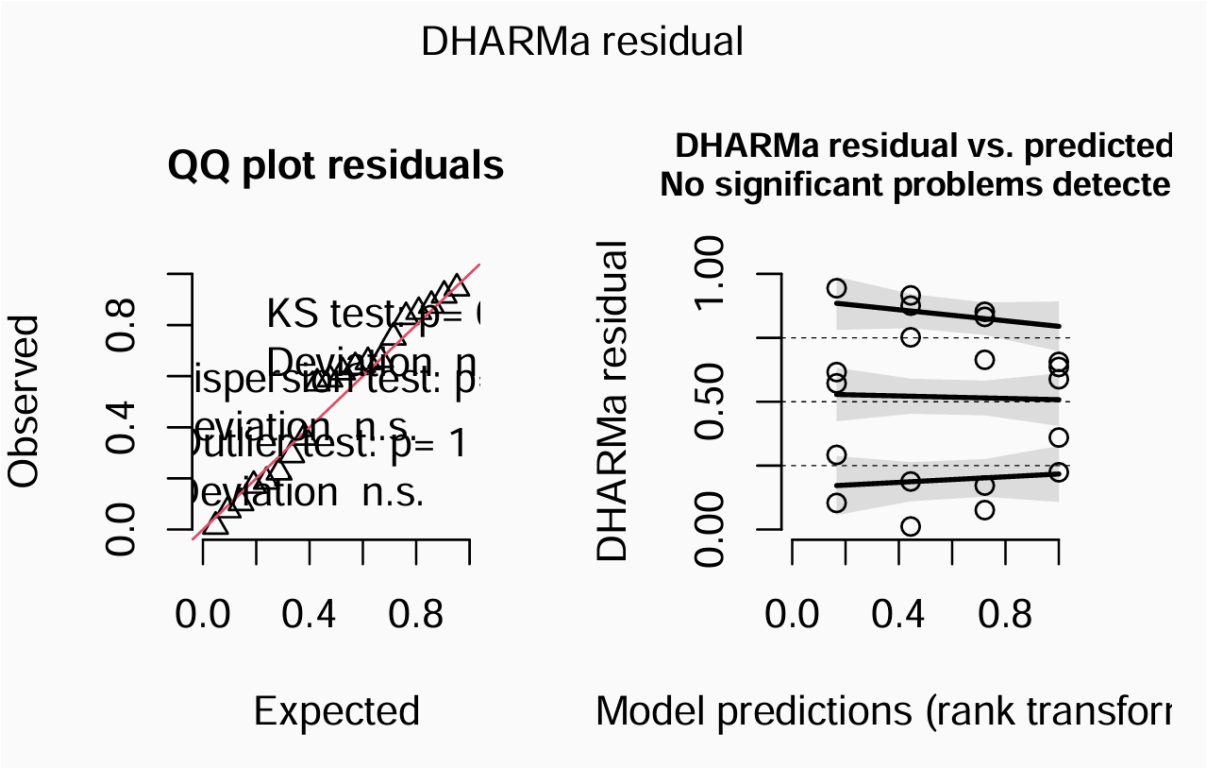
7.1 Descriptive table with means, sample size (n), standard deviation (SD), and standard error (SE) for mortality over all laboratory test (count).

| Group | Mean | n | SD | SE |
|---|---|---|---|---|
| CTR | 2.8 | 5 | 1.4832 | 0.6633 |
| DEL | 3.4 | 5 | 1.8166 | 0.8124 |
| IMI | 4.4 | 5 | 0.8944 | 0.4000 |
| COM | 3.2 | 5 | 2.1679 | 0.9695 |

5.2 Model selection tables based on Akaike Information Criteria (AIC) for mortality

| Mod_name | Family | Random | df | AICc | Delta |
|---|---|---|---|---|---|
| mod_mortalidade.1 | gaussian (id) | | 5 | 86.8 | 0.00 |
| mod_mortalidade.2 | negative binomial2 (lg) | | 5 | 90.0 | 3.20 |
| mod_mortalidade | gaussian (id) | | 8 | 97.9 | 11.11 |

4.3 Best model residuals: mod_mortalidade.1




4.4 Best model for syrup consumption — mod_consumo.g (gaussian, identity) — summary

| | **Estimate (log)** | **std.error** | **z value** | **p value** |
|---|---|---|---|---|
| Intercept(CTR) | 2.8000 | 0.6633 | 4.221 | $2.43.10^{-05}$ |
| DEL | 0.6000 | 0.9381 | 0.640 | 0.5224 |
| IMI | 1.6000 | 0.9381 | 1.706 | 0.0881 . |
| COM | 0.4000 | 0.9381 | 0.426 | 0.6698 |